ECON100B - SECTION 8

The Romer Model

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ANNOUNCEMENTS/AGENDA

Announcements

- PSET 2 due Friday, 2/27
- In-Class Midterm, 3/5

Agenda

- Questions from PSET 1?
- Romer Model

Question of the Day

1. PSET 2, Question 1.a: Recall that real GDP measured on the expenditure side is given by: $Y = C + I + G + NX$. What happens if there is no government and no foreign trade? Rewrite the equation and briefly explain what it states about the disposition of income, Y. Does a component of income become interpretable as something else? Explain.
2. PSET 2, Question 1.b: In the U.S., what can you say about the level of the long-run share of consumption in GDP, meaning C/Y? Briefly discuss.
3. What is your favorite movie of all time?

PSET 1 Takeaways

1. Short term economic shocks that don’t change any parameters in a model don’t shift curves but can cause movements in a model. Long term economic shocks that change parameters in a model may shift curves.
2. Be careful to read and complete all parts of a problem – if a questions asks you to illustrate on a graph and show your work/discuss your answer, you must do both!
3. Assign your problems to pages ☺

The Romer Model

What kinds of economic goods are there?

- We've covered physical objects, including items for consumptions (phones, pencils, clothes), items for capital (machinery, factories, buildings), and human labor

- In the Solow model we saw a huge part of economic output attributed to "A" or a Total Factor Productivity parameter

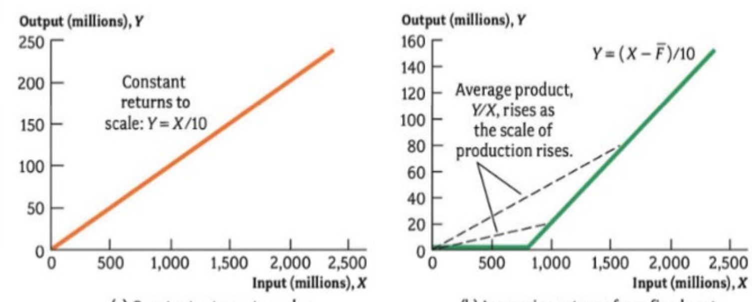
- The Romer model introduces us to Ideas
    - How to invent an object? How to turn mundane objects into complex objects?
    - How to manage a company or a supply chain?
    - How to tackle a complex and unprecedented challenge?

Economics of Ideas

- Ideas: Instructions that tell us how we can arrange and work with raw materials in ways that are economically useful
    - Sustained economic growth occurs because we discover new ideas
- Nonrival goods: A good that does not reduce usefulness when used by a particular person
    - Consider rival goods: phones, pencils, clothes, most goods in the economy! My use diminishes/inhibits your ability to use the good
    - Nonrivalrous goods are goods that don’t exhibit this behavior. For example, my use of algebra doesn’t diminish your ability to use algebra. You singing Trap Queen doesn’t reduce my ability to sign Trap Queen.
    - A nonrival good can still be excludable – excludability refers to the extent to which someone has property rights over a good/idea

Economics of Ideas

- Increasing Returns refers to the idea that the output of a good has increasing returns to scale as we increase the amount of inputs
  - For example, a new antibiotic may be extremely expensive to produce at first because of the significant costs of research and development, but becomes cheaper to produce once the R&D is done and production increases



(a) Constant returns to scale: $Y = X/10$.

(b) Increasing returns from fixed cost: $\bar{F} = 800$.

A New Production Function

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{1/3} L_t^{2/3}.$$

- Let’s introduce a new production function, one that replaces our TFP parameters from Cobb Douglas/Solow with a new input called A, which represents “knowledge”/stock of ideas
- Since we can now change/scale our parameter A, we notice that our production function no longer has constant returns to scale, but rather increasing returns to scale
    - We still have constant returns to objects in production, so if we increase our ideas as well, we will have increasing returns to scale

Pure Competition

- Under the idea of perfect competition, markets lead to Pareto Optimal outputs (i.e. no way to change the allocation to make someone better off without making someone else worse off)
- Prices equate to marginal costs and marginal benefits
- Say a pharmaceutical company sells a drug at the marginal cost, due to perfect competition
    - If they had to invest in R&D expenditures as well, they would lose money if selling at marginal cost
    - Meanwhile, other competitors can sell at the marginal cost and break even due to the "idea" of the drug as nonrivalrous
- This helps explain the purpose of the patent and copyright systems (and trade secrets)
- Governments, altruism, prizes, and other approaches can help address this problem of the "cost" of ideas
- No right/best approach to incentivize/compensate for the benefits of innovation

Practice: Ideas in Economics

1. Is a computer a rival or nonrival good? What about the design for a computer?
2. Under our new Production Model,

$$Y = F(A, K, L) = A_t K_t^{1/3} L_t^{2/3}$$

What happens to output when we triple capital and labor? What happens to output when we triple capital, labor, and ideas (A)?

3. What is the economic justification for implementing patents? What other tools can be put in place to incentivize innovation/generation of new ideas?

Practice Solutions

1. A computer is a rival good. If one person is using it, another person cannot at the same time. The idea for the design of a computer is nonrivalrous. A factory can make multiple of the same computers using the design at the same time
2. If we triple capital and labor, we triple output (CRS). If we triple capital and labor in addition to tripling ideas, our output increases by 9x (!)
3. Patents are economically justified because they help compensate for the costs of innovation/ideas. Government spending, private investment, and altruistic competition can be seen as other ways to economically incentivize innovation

The Romer Model - Setup

- Consider the pizza model that we discussed previously with the Cobb Douglas and Solow economies
- Under Solow, growth stopped once we reached a certain level of capital
- Under the Romer Model, consider the fact that we can have ideas contributing to increases in output, allowing us to sustain growth!

The Romer Model Setup

- For now, lets ignore capital and just focus on Ideas and Labor

$$Y_t = A_t L_{yt}$$
$$\Delta A_t = \bar{z} A_t L_{at}.$$

- Output is produced using the stock of existing knowledge and labor
- The change in the stock of ideas is the "number of new ideas" produced in time period t.
- Ideas are produced using existing ideas and workers, multiplied by a parameter "z", which represents how good our economy is at producing ideas. Workers can either produce output (y) or ideas (a)

$$L_{yt} + L_{at} = \bar{N}.$$

The Romer Model – Setup

- How do we determine how much labor goes toward innovation vs. output?
- In this class, we're given the parameter exogenously:

$$L_{at} = \bar{l}\bar{N}$$

- Food for thought: In a market, the market will naturally try to find the optimal point of innovation. What would happen to our parameter l under an unregulated free market economy? How about an economy that has patents and protections for innovation?

Solving the Romer Model

- Output per person can be represented as:

$$y_t \equiv \frac{Y_t}{\bar{N}} = A_t(1 - \bar{\ell}).$$

- We can see that output per person depends on the total stock of knowledge (nonrivalry) as opposed to per-capital stock as we saw in the Solow model

- We can find our growth in ideas with the following:

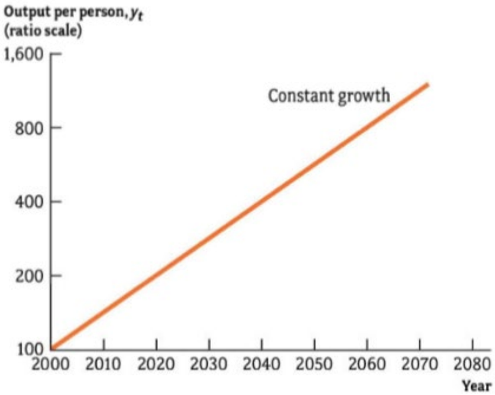
$$\frac{\Delta A_t}{A_t} = \bar{z}L_{at} = \bar{z}\bar{\ell}\bar{N}.$$

$$A_t = \bar{A}_0(1 + \bar{g})^t.$$

Solving the Romer Model

Our final Romer model output per person equation is:

$$y_t = \overline{A_0}\left(1 - \bar{l}\right)(1 + \bar{g})^t$$



Practice with the Romer Model

1. What would happen to the growth rate of output per person if we increased the number of people in the population (N)? What does this tell us?
2. What would happen to the growth rate of output per person if we increased the share of people that were working in research (l)? What does this tell us?

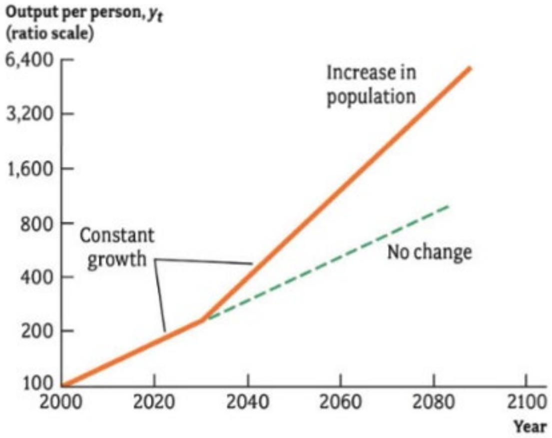
3. Suppose we're given a Romer model with the following parameters:
$\bar{z} = 0.01, \bar{l} = 0.05, \bar{N} = 100, \overline{A_0} = 1000$

What is our output per person in 10 years? What is our output per person in 20 years? In 100 years?

Recall: $y_t = \overline{A_0}(1 - \bar{l})(1 + \bar{g})^t$ and $\bar{g} = \bar{z}\bar{l}\bar{N}$

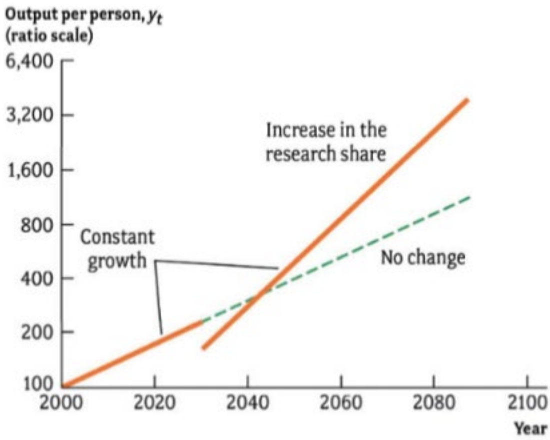
Practice Solutions

Output per Person after an Increase in $\bar{N}$



Practice Solutions

Output per Person after an Increase in ℓ



Practice Solutions

3. First calculate the growth rate g: 0.01 * 0.05 * 100 = 0.05

Then plug into our output per person formula to find:

10 year output per person: 1,547
20 year output per person: 2,520.63
100 year output per person: 124,926.195

PSET Questions

(c) [2 points] Consumption goods and services include things like food and health care. Investment goods and services include container cranes for seaports, architectural services, and the office buildings themselves. Do the same workers in the economy typically produce both consumption goods and investment goods? Explain.

(d) [2 points] Given your previous answers, what should be true in the short run about the U.S. share of labor engaged in producing consumption goods, if consumption and investment sectors require and benefit from the same amount of capital and technology?